

# Index-Linked / CMS Channel Note Pricer

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model: "index_linked"
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*Inflation-index-linked structured channel notes*

## Description

Prices structured notes where both coupons and redemption are linked to an inflation index ratio. The index ratio is projected forward from a known current level using a flat annual inflation assumption. Designed for structured retail products with a capital-protection element and an index-linked upside, such as inflation-linked channel notes issued by banks.

## Pricing Methodology

The index ratio at any future date is projected as:

$$\text{IndexRatio}(d) = \text{index\_ratio\_at\_eval} \times (1 + \text{annual\_index\_growth\_rate})^t$$

Each structured coupon is multiplied by the index ratio at its payment date.

## Key Formulas

$$\text{IndexRatio}(d) = \text{IR\_eval} \times (1 + g)^{\text{yearFraction}(\text{eval}, d)}$$
$$\text{coupon}(t) = \text{coupon\_multiplier} \times \text{IR}(t) \times \text{notional} \times \text{accrual}$$
$$\text{Note PV} = \text{Collateral PV} + \text{indexed\_swap\_adjustment} - \text{issuer\_spread\_PV}$$

## Required Fields

| Field                                            | Type   | Description                                |
|--------------------------------------------------|--------|--------------------------------------------|
| instrument_id                                    | string | ISIN or internal identifier                |
| evaluation_date                                  | string | Pricing date                               |
| coupon_structure                                 | string | Must be "index_linked"                     |
| note_notional                                    | float  | Notional of the note                       |
| collateral                                       | object | Collateral bond definition (same as SPIRE) |
| index_linked_assumption                          | object | Container for inflation assumptions        |
| index_linked_assumption.index_ratio_at_eval      | float  | Current index ratio                        |
| index_linked_assumption.annual_index_growth_rate | float  | Forward inflation rate                     |
| index_linked_assumption.coupon_multiplier        | float  | Real coupon multiplier                     |

## Optional Fields

| Field                | Type   | Description                          |
|----------------------|--------|--------------------------------------|
| credit_spread_bp     | float  | Issuer spread (default 0)            |
| collateral_spread_bp | float  | Collateral leg spread (default 0)    |
| issuer_call          | string | "Applicable" to enable call schedule |
| call_dates           | list   | List of call dates (DD-MM-YYYY)      |

## Key Outputs

- selected\_npv — index-linked note price
- npv\_to\_maturity, npv\_to\_first\_call, npv\_to\_worst\_call
- index\_ratio\_at\_maturity — projected final index ratio
- cashflows — per-period including index-adjusted coupons